Adversarial Complementary Learning for Multisource Remote Sensing Classification

Yunhao Gao, Mengmeng Zhang, Wei Li, *Senior Member, IEEE*, Xiukai Song, Xiangyang Jiang, and Yuanqing Ma

***Abstract*— Convolutional neural networks (CNNs) have attracted increasing attention in the field of multimodal cooperation. Recently, the adoption of CNN-based methods has achieved remarkable performance in multisource remote sensing data classification. However, it is still confronted with challenges in the aspect of complementarity extraction. In this article, the adversarial complementary learning (ACL) strategy is embedded into the CNN model called ACL-CNN, which is employed to extract the complementary information of the multisource data. The proposed ACL-CNN is able to filter out the common patterns and specific patterns from multisource data by conducting the adversarial max–min game. Especially, the modality-independent common patterns constitute the basic representation of the land covers, while the specific patterns are linearly independent of the common patterns that provide the supplementary representation. Therefore, the complementary information is mapped to a compact and discriminative representation. To eliminate the singularity noise, a learnable pattern sampling module (PSM) is designed to extract the mutual-exclusion relationship between specific patterns. Extensive experiments over three datasets demonstrate the superiority of the proposed ACL-CNN compared with several classification technologies.**

***Index Terms*— Adversarial complementary learning (ACL), adversarial max–min game, convolutional neural network (CNN), multisource remote sensing classification, pattern sampling module (PSM).**

I. Introduction

THE recent success of Earth observation has been attributed to the developments of remote sensing. Remote sensing data exhibits spatial, spectral, and polarization properties depending on the type of employed sensor and measured radiation [1]. Among them, hyperspectral images (HSIs) deserve special attention and provide highly relevant information ranging from the visible spectrum to the infrared spectrum, which promotes satisfying classification results [2]. However, the resolution of spatial components is limited by the requirements of signal-to-noise ratio. Fortunately, multisource remote sensing makes omnidirectional observation possible [3], [4]. A synthetic aperture radar (SAR) is a coherent imaging radar system with high resolution, which has the capabilities of all-weather and all-time observation [5]. Besides, the multispectral images (MSIs) and light detection and ranging (LiDAR) have the advantage of representing morphological and elevation information, respectively, which are complementary to HSIs. Therefore, how to completely fuse multisource remote sensing data becomes a crucial issue.

A common approach to integrating multisource information is data fusion. Data fusion converges the advantages of different data sources to reconstruct higher quality images [6]. In [7], the input images were unmixed into the pure reflectance and the associated mixing coefficients, and thus, the data fusion was achieved. Akhtar et al. [8] proposed the nonparametric Bayesian sparse representation for HSI super-resolution. Zhang and Prasad [9] proposed a robust formulation for data fusion, in which the sparse coefficients are constrained by an adaptive locality weight. Dong et al. [10] jointly estimated the high-resolution HSIs by hyperspectral dictionary and sparse codes based on the spatial–spectral sparse prior. In [11], the HSI and MSI were fused via coupled sparse tensor factorization. After that, the subspace-based low tensor multirank regularization is developed for data fusion [12]. In [13], the HSI and MSI are fused via a deep convolutional neural network (CNN) that incorporates the learned deep priors. However, data fusion is more suitable for improving the data quality of homogeneous sensors, such as HSI/MSI, MSI/panchromatic image, and so on. Its poor applicability to heterogeneous data easily leads to information loss.

Because of the success of feature engineering in image interpretation [14], [15], researchers have carried out a large number of experiments on collaborative classification by feature fusion. For feature fusion, the features of different data sources are jointly transformed into high-dimensional space, and then, the classifier is designed for land-covers recognition. The support vector machine (SVM) and extreme learning machine (ELM) stand out for their efficiency in land-covers classification. In [16], the projected multisource features from HSI and LiDAR data were jointly classified by SVM. Recently, extinction profiles (EPs) have been widely utilized to represent spatial characteristics, which have proved to be helpful for remote sensing data classification [17], [18]. In [19], the EPs-based spatial feature and the spectral feature were fused by the sparse and low-rank component analysis. In [20], EPs

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are employed to extract the spatial and elevation information from HSI and LiDAR, and then, the features were fused in a lower dimensional space by orthogonal total variation component analysis. To emphasize the correlation of heterogeneous features, multiple-kernel learning (MKL) naturally establishes the linear combinations among multiple basis kernels to learn the optimal one. In [21], the suboptimal kernels were selected by kernel alignment (KA), and then, MKL was carried out to explore the optimal combination of spatial–spectral and elevation attributes. Zhang et al. [22] combined the active learning (AL) and MKL, named ensemble MKL-AL to identify the additional informative and representative samples from HSI and LiDAR data. Besides, some of the most noteworthy approaches also achieve the promising performance, such as graph-based method [23], random forest [24], manifold alignment [25], and segmentation [26]. However, the successful adoption of the above methods depends on extensive domain expertise and prior knowledge, which are time-consuming and laborious.

Since Krizhevsky et al. [27] adopted CNN for image recognition, different CNN-based extensions have achieved excellent performance in detection, segmentation, and natural language processing [28], [29], [30], [31]. More recently, the CNN-based methods have made lots of attempts in multisource image classification, which also achieved the promising results [32], [33]. For instance, Ghamisi et al. [34] utilized a CNN to classify the stacked the EPs and spectral features. Later, Chen et al. [35] proposed an end-to-end CNN, in which the extracted features from multi-/hyperspectral and LiDAR data were fused by the fully connected (FC) layers. In [36], the feature extractors were designed specifically for different data sources. In [37], the CNNs were coupled to learn from each other, and then, the maximization and weighted summation strategies were conducted for feature fusion and decision fusion, respectively. More recently, some feature interaction methods have been widely used in collaborative classification. In [38], the latent representation of different convolutional layers interacted for a joint representation. In [39], the interleaving perception CNN was designed to integrate the multisource features. Gao et al. [40] designed a depth cross attention module (CAM) to emphasize the correlation. Li et al. [41] constructed a graph-based network, in which the transition graph was utilized to enhance the association of HSI and MSI. In [42], a multiscale interactive fusion network was developed to fuse the multiscale information. Considering the information heterogeneity and imbalance of multisource remote sensing data, the asymmetric feature fusion scheme was proposed. In [43], the parameters of independent batch normalization (BN) layers are used to control feature propagation. In [44], the asymmetric fusion was achieved by shifted feature fusion and feature selection. Zhang et al. [45] developed a fusion classification scheme based on structural optimization transmission. Li et al. [46] designed a triplet semisupervised deep network for HSI and LiDAR data classification.

Although these aforementioned CNN-based approaches have achieved promising performance, they still suffer from the following issues. First, *redundant feature information*—as shown in Fig. 1(a), traditional methods are difficult to filter out different feature patterns, which inevitably leads to information redundancy [25], [47]. Therefore, it is urgent to construct complementary feature by mining common and specific patterns, as shown in Fig. 1(b). Second, *interferential singularity noise*—as shown in Fig. 1(b), the common patterns and specific patterns express the basic representation and supplementary representation of land covers, respectively. In fact, the supplementary representation suffers from the interferential singularity noise (redundant spectrum and speckle noise), which weakens the complementarity of different pattern stacking as well as the classification performance. Therefore, it is necessary to explore the mutually exclusive relationships between specific patterns.

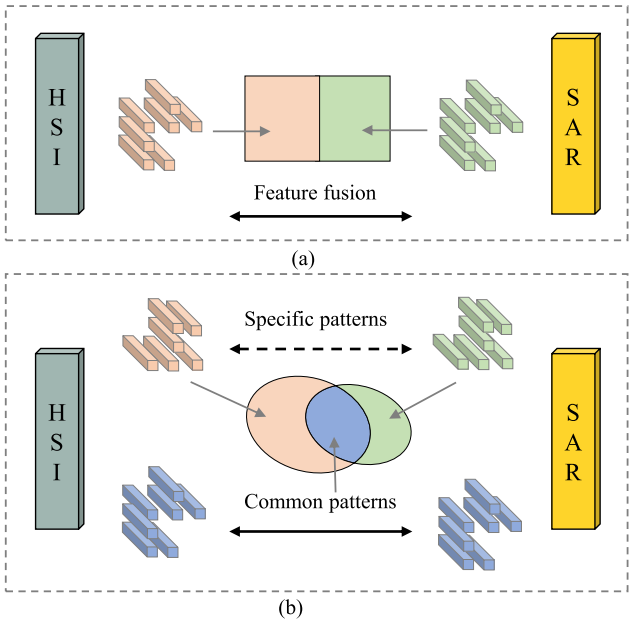

Fig. 1. Illustration of the difference between (a) traditional feature fusion and (b) complementary feature construction.

To tackle the above issues, a CNN based on adversarial complementary learning (ACL-CNN) is proposed to collaboratively classify multisource images. First, four groups of encoders with the same structure are employed for feature embedding, in which two groups of encoders with shared weight are utilized as common encoders to extract the common patterns. In addition, the other two groups of encoders with independent parameters are the specific encoders, which is utilized to extract the specific patterns. In the discrimination stage, the max–min adversarial strategy drives the model to accurately distinguish the modality labels, while the specific patterns are correlated with the common patterns. In the generation stage, the common patterns are irrelevant to the modality, while the specific patterns are linear independence of common patterns. To explore the meaningful information of specific patterns in the classification task so as to maximize revenue, the pattern sampling module (PSM) is designed to weighted integrate the specific patterns for classification. Extensive experiments with different configurations have been conducted using three multisource datasets to verify the effectiveness. The obtained results demonstrate that the proposed ACL-CNN achieves superior performance advantages over the other relevant classification methods.

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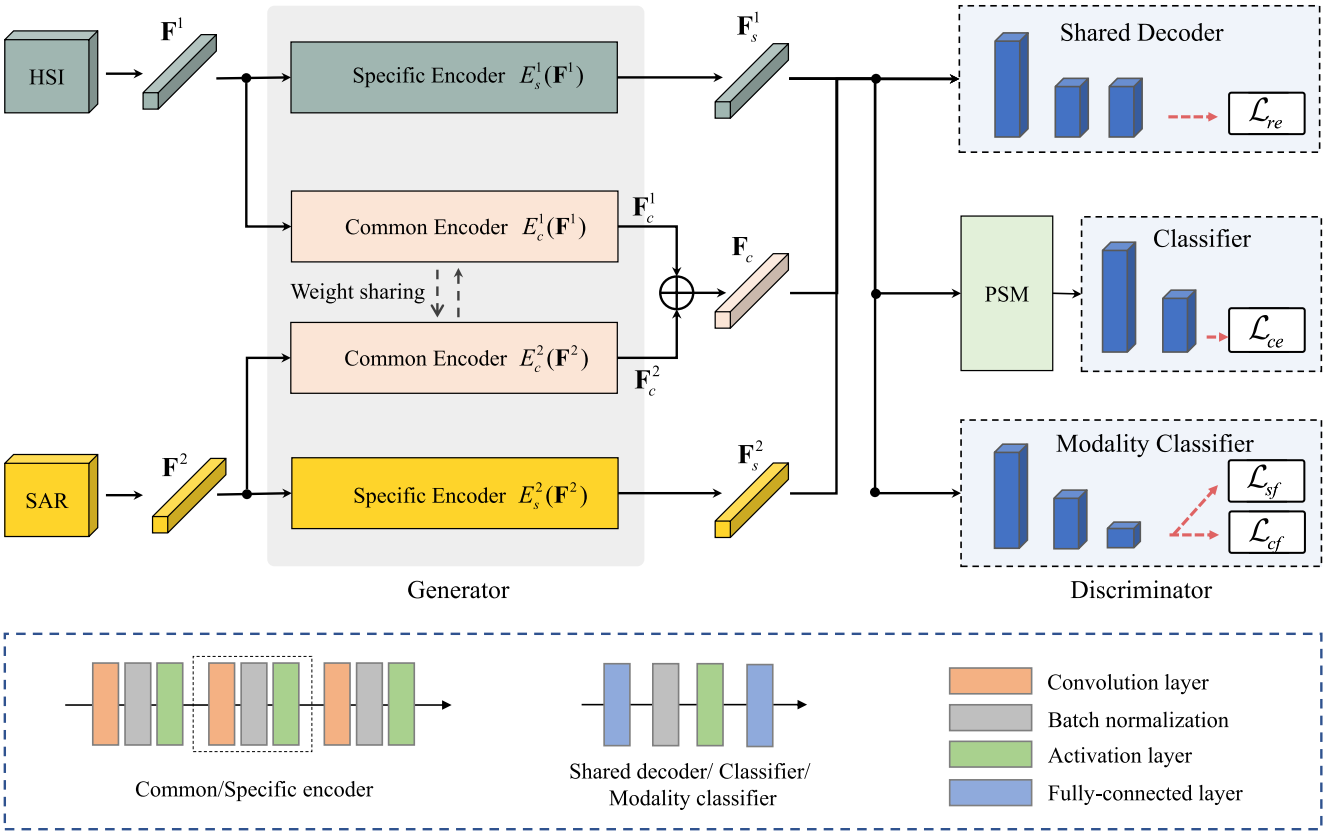

Fig. 2. Framework of the proposed ACL-CNN. The common patterns and specific patterns of multisource data are generated by the adversarial generator and discriminator, and then, the PSM is utilized for the selection of specific patterns. Finally, the concatenated common and specific patterns are fed into the classifier for label prediction.

The main contributions of the proposed ACL-CNN are summarized as follows.

1) An ACL-CNN is developed to mine the complementary merits of multisource remote sensing data for land-covers classification. The common and specific encoders optimized by the complementary learning strategy are utilized to extract the common patterns and specific patterns between different data sources. After that, the specific patterns are weighted integrated to achieve the complementary merits.
2) On the one hand, the common patterns are fed into the modality classifier to identify its modality labels by the cross-entropy (CE) loss. On the other hand, the specific patterns are constrained by the mean square error (mse) loss to ensure its linear independence of common patterns. The ACL strategy is conducted by minimizing the losses in the generation stage and maximizing in the discrimination stage. Therefore, the modality-independent common patterns and complementary specific patterns are generated in the generation stage.
3) To eliminate the singular noise, a PSM is designed to extract the mutually exclusive relationship between the specific patterns. The meaningful information is explicitly emphasized, while the unnecessary one is suppressed. Therefore, the problem of singularity noise is alleviated.

The remainder of this article is organized as follows. The proposed classification framework is described in Section II. Section III validates the performance of the proposed method by providing the discussion of parameter configuration and comparison with related works. Finally, Section IV concludes this article with some remarks and directs future research lines.

II. Proposed Classification Framework

The coregistered multisource remote sensing data captured over the same geographical region are represented as $[X_1, X_2, \ldots, X_m, Y]$. Here, $[X_1, X_2, \ldots, X_m]$ are the patch pairs centered at the corresponding pixels of the multisource data, m is the number of the modalities, and Y is the labels.

As illustrated in Fig. 2, the patch pairs are first fed into the generator to obtain the embedding features, and then, the discriminator is utilized to filter out the common and specific patterns by the adversarial strategy. At last, the PSM is designed to balance the contribution of different specific patterns. Therefore, the ACL is achieved, which is described as follows.

A. Encoder Network and Modality Classifier for Complementary Learning

Generally speaking, the common and specific patterns of multisource remote sensing data constitute the compact and discriminative feature representation. However, the information overlap of common–specific patterns deteriorates the classification performance. Specifically, the common patterns containing specific information lead to information redundancy, whereas special patterns containing common patterns

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reduce the complementarity. Therefore, this article designs a generator (encoder) and a discriminator (modality classifier) to filter out the common and specific patterns.

1) Encoder Network: Taking the HSI and SAR images as examples, the encoders are designed for feature embedding. Concretely, two independently optimized specific encoders are utilized to extract the specific patterns, and the other two common encoders with shared weights are utilized to extract the common patterns of different modalities as follows:

$$\begin{cases} \mathbf{F}_s^1 = E_s^1(\mathbf{F}^1) = E_s^1(\text{Conv}(\mathbf{x}_1)) \\ \mathbf{F}_s^2 = E_s^2(\mathbf{F}^2) = E_s^2(\text{Conv}(\mathbf{x}_2)) \\ \mathbf{F}_c^1 = E_c^1(\mathbf{F}^1) = E_c^1(\text{Conv}(\mathbf{x}_1)) \\ \mathbf{F}_c^2 = E_c^2(\mathbf{F}^2) = E_c^2(\text{Conv}(\mathbf{x}_2)) \end{cases} \tag{1}$$

where $\text{Conv}(\cdot)$ is denoted as a convolutional operator, which is employed to match the dimension between $\mathbf{x}_1$ and $\mathbf{x}_2$. $\mathbf{x}_1, \mathbf{x}_2 \in [X_1, X_2]$ denotes a patch pair. $\mathbf{F}_s^1$ and $\mathbf{F}_s^2$ are the specific patterns of the HSI and SAR image, respectively. $\mathbf{F}_c^1$ and $\mathbf{F}_c^2$ are the common patterns. E_s^1, E_s^2, E_c^1, and E_c^2 are the encoder networks with the same structure. The encoder network is designed as $\text{ConvBN}(d, d) \rightarrow \text{ConvBN}(d, d) \rightarrow \text{ConvBN}(d, d)$. Here, b denotes the dimension of common/specific patterns. ConvBN consists of a convolution layer, followed by a BN layer and *ReLU* function.

2) Modality Classifier: To generate the modality-independent common patterns, the adversarial learning strategy has been practiced and proved effective [48], [49]. The modality classifier is performed as a discriminator, which is utilized to judge whether common patterns from common encoders belong to the same modality. The modality classifier is designed as $FcBN(d, d) \rightarrow FC(d, m)$. Here, m denotes the number of modalities. $FcBN$ consists of an FC layer, followed by a BN layer and *ReLU* function.

The CE loss is employed to optimize the modality classifier as follows:

$$\mathcal{L}_{cf} = -\frac{1}{N}\sum_{j=1}^{N}\left(y_m \cdot \left(\log p\left(\mathbf{F}_c^1, \Theta_D\right) + \log p\left(\mathbf{F}_c^2, \Theta_D\right)\right)\right) \tag{2}$$

where N is the batch size, and y_m is the one-hot encoding of the modality labels. The modality label of HSI is $[1, 0]$, while the label of SAR modality is $[0, 1]$. $p(\cdot, \Theta_D)$ denotes the prediction probability by modality classifier Θ_D. In the generation stage, the encoder networks fool the modality classifier by generating modality-independent common patterns, while in the discrimination stage, the modality classifier tries to correctly distinguish the modal labels of common patterns. Therefore, the multisource common patterns are modality-independent through the max–min game.

To extract the specific patterns of multisource data, which is complementary to the common patterns, the mse loss is employed to constrain the linear independence as follows:

$$\mathcal{L}_{sf} = \frac{1}{N}\sum_{j=1}^{N}\left(\left\|\mathbf{F}_c^1 - \mathbf{F}_s^1\right\|_2 + \left\|\mathbf{F}_c^2 - \mathbf{F}_s^2\right\|_2\right) \tag{3}$$

where $\|\cdot\|_2$ denotes L_2-norm. Similarly, in the generation stage, the specific encoder tries to generate specific patterns that are uncorrelated to common patterns. In the discrimination stage, we try to make the common patterns align with the specific patterns. Therefore, the specific patterns of different modalities are generated.

Generally speaking, the adversarial learning strategy is employed to filters out the common patterns $[\mathbf{F}_c^1, \mathbf{F}_c^2]$ and specific patterns $[\mathbf{F}_s^1, \mathbf{F}_s^2]$ from multisource data. The common patterns are further enhanced by summation as follows:

$$\mathbf{F}_c = a \cdot \mathbf{F}_c^1 + b \cdot \mathbf{F}_c^2, \quad a + b = 1 \tag{4}$$

where a and b are the weights generated by $\delta(W_c)$. $\delta(\cdot)$ is the *softmax* function, and $W_c \in \mathbb{R}^{2\times 1}$ is the weight matrix.

B. Shared Decoder for Feature Integrity

The common encoders and the specific encoders mentioned above have the same structure; however, they are optimized with different parameters. Therefore, the common patterns and specific patterns naturally maintain linear independence, which is easy to cause shortcut optimization. To drive the model learning, the shared decoder is designed as $FcBN(2d, d) \rightarrow FC(d, d)$, which is calculated for feature reconstruction as follows:

$$\begin{cases} \hat{\mathbf{F}}^1 = \mathbf{W}_{d2} \cdot \sigma\left(\mathbf{W}_{d1}\left(\text{concat}\left(\mathbf{F}_c, \mathbf{F}_s^1\right)\right)\right) \\ \hat{\mathbf{F}}^2 = \mathbf{W}_{d2} \cdot \sigma\left(\mathbf{W}_{d1}\left(\text{concat}\left(\mathbf{F}_c, \mathbf{F}_s^2\right)\right)\right) \end{cases} \tag{5}$$

where $\hat{\mathbf{F}}^1$ and $\hat{\mathbf{F}}^2$ are the reconstructed features. $\mathbf{W}_{d1}$ and $\mathbf{W}_{d2}$ denote the weight matrix of two FC layers in shared decoder. $\text{concat}(\cdot)$ is the concatenation operator along the channel dimension, and $\sigma(\cdot)$ is the *ReLU* function. The quality of reconstructed features is evaluated by mse loss as follows:

$$\mathcal{L}_{re} = \frac{1}{N}\sum_{j=1}^{N}\left(\left\|\hat{\mathbf{F}}^1 - \mathbf{F}^1\right\|_2 + \left\|\hat{\mathbf{F}}^2 - \mathbf{F}^2\right\|_2\right) \tag{6}$$

where $\hat{\mathbf{F}}^1$ and $\hat{\mathbf{F}}^2$ are the reconstructed features. Therefore, the feature integrity is guaranteed, and the common and specific patterns are guided into complementary merits.

C. PSM and Classifier for Feature Integration

1) Pattern Sampling Module: The common and specific patterns have the ability to represent the data characteristics in a targeted manner. The common patterns are the basic representation of land covers, while specific patterns are utilized and used as their complementary representation. However, some redundant spectrum and speckle noise are mixed in the specific patterns, which provide less contribution or even negative gain for classification. Therefore, a PSM, as shown in Fig. 3, is designed for reweighing the distribution of different specific patterns. First, the specific patterns are concatenated along the height dimension as follows:

$$\mathbf{F}_s^* = \text{concat}^*\left(\mathbf{F}_s^1, \mathbf{F}_s^2, \ldots, \mathbf{F}_s^m\right) \tag{7}$$

where $\mathbf{F}_s^* \in \mathbb{R}^{c\times m\times 1}$ is the concatenated feature. c is the number of channels, and m is the number of modalities. $\text{concat}^*(\cdot)$ is the concatenation operator along the height dimension.

A learnable sampling method is used to select the meaningful information from different specific patterns, rather than

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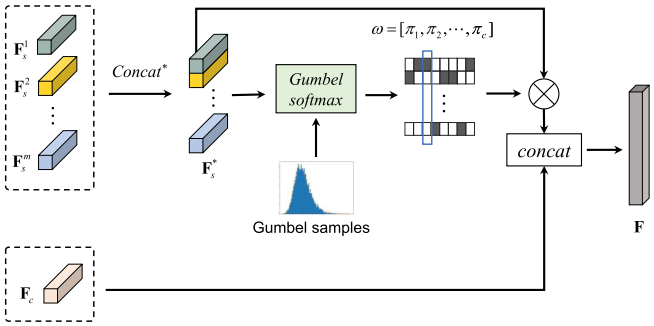

Fig. 3. Illustration of the PSM.

reweighting each pattern through an attention vector. The feature index obtained by sampling method is denoted as follows:

$$\omega = \text{sample}\left(\mathbf{F}_s^*\right) \tag{8}$$

where $\omega = [\pi_1, \pi_2, \ldots, \pi_c], \pi = \{0, 1\}^{m\times 1}$. The sampling weight is a discrete variable that represents the mutually exclusive relationship (0 or 1) of each channel position. Considering the optimization of mutually exclusive relationship, the Gumbel-softmax trick is employed to reparameterize the discrete variables [50]. Concretely, ω is denoted as follows:

$$\omega = \text{one_hot}\left(\arg\max_i\left(\log\left(\omega^i\right) + \epsilon^i\right)\right) \tag{9}$$

where ω^i is the class probability, ϵ^i is the random variable, and $i \in \{1, 2, \ldots, m\}$. The Gumbel distribution is a continuous distribution on the simplex, which can be sampled to generate the random variable ϵ. ϵ is denoted as follows:

$$\epsilon = -\log(-\log(u)), \quad u \sim \text{Uniform}(0, 1). \tag{10}$$

In practice, the *argmax*$(\cdot)$ operation is not differential. Therefore, the continuous and differentiable approximation of argmax is achieved by *softmax* function as follows:

$$h^* = \frac{\exp\left(\left(\log\left(\omega^i\right) + \epsilon^i\right)/\tau\right)}{\sum_{i\in\{1,2,\ldots,m\}}\exp\left(\left(\log\left(\omega^i\right) + \epsilon^i\right)/\tau\right)} \tag{11}$$

where $\tau > 0$ is the temperature parameter. When the temperature τ approaches 0, the samples from the Gumbel-softmax distribution is one hot and identical to the categorical distribution. Therefore, the sampled specific patterns are calculated as follows:

$$\mathbf{F}_s^s = \text{hsum}\left(\omega \otimes \mathbf{F}_s^*\right) \tag{12}$$

where $\mathbf{F}_s^s \in \mathbb{R}^{c\times m\times 1}$ is the sampled specific pattern. hsum$(\cdot)$ is the mean operator along the height dimension.

Finally, the output feature is generated by concatenating the common patterns and the resampled specific patterns as follows:

$$\mathbf{F} = \text{concat}\left(\mathbf{F}_c, \mathbf{F}_s^s\right) \tag{13}$$

where $\mathbf{F} \in \mathbb{R}^{2c\times 1\times 1}$ denotes the output feature.

2) Classifier: For land-covers classification, the classifier is utilized to predict the labels, which is designed as $FcBN(2d, d) \rightarrow FC(d, C)$. Here, C denotes the number of classes. The probability distribution is generated by receiving the output feature $\mathbf{F}$. To distinguish the features of different classes, the CE loss is utilized for optimization as follows:

$$\mathcal{L}_{ce} = -\frac{1}{N}\sum_{j=1}^{N} y \cdot \log p(\mathbf{F}, \Theta_C) \tag{14}$$

where y is the one-hot encoding of the class labels, and $p(\cdot, \Theta_C)$ is the prediction probability by the classifier Θ_C.

D. Analysis of Optimization Algorithm and the ACL-CNN

In this article, the proposed method is end-to-end optimized with ACL strategy. Especially, the generator (encoder) and the discriminator (modality classifier) run as a max–min game that targets beating each other at optimization. The shared decoder and classifier are utilized for feature integrity and label prediction, respectively. Therefore, the generation loss $\mathcal{L}_{\text{gen}}$ and the discrimination loss $\mathcal{L}_{\text{dis}}$ are defined as follows:

$$\begin{aligned}\mathcal{L}_{\text{gen}} &= \mathcal{L}_{ce} + \mathcal{L}_{re}\\ \mathcal{L}_{\text{dis}} &= \mathcal{L}_{cf} + \mathcal{L}_{sf}.\end{aligned} \tag{15}$$

The adversarial learning strategy is achieved by alternately minimizing and maximizing the total loss as follows:

$$\begin{cases}\mathcal{L} = \arg\min\left(\mathcal{L}_{\text{gen}} - \lambda\cdot\mathcal{L}_{\text{dis}}\right), & \text{if} \quad E\%I \neq 0\\ \mathcal{L} = \arg\max\left(\mu\cdot\left(\mathcal{L}_{\text{gen}} - \lambda\cdot\mathcal{L}_{\text{dis}}\right)\right), & \text{else}\end{cases} \tag{16}$$

where $\mathcal{L}$ is the total loss, and λ is the hyperparameter to balance the weight between generation loss and discrimination loss. Empirically, the discrimination loss is easier to converge in the optimization process. Therefore, an interval $I = 5$ is utilized to adjust the alternating training of the generator and discriminator. That is to say, the discriminator is optimized once every I epochs. $\mu = 1.0$ is set to adjust the weight of the discrimination stage. The stochastic gradient descent (SGD) algorithm is applied to train the whole algorithm in 300 epochs. All the experiments are implemented on the Pytorch 1.9.0 software platform. The hardware environment consists of Intel Core i7-9700 @ 3.00-GHz CPU, 40 GB of DDR4, and Nvidia GTX 3070Ti GPU.

Figs. 4 and 5 display the relationship between the training loss and the number of epochs. From the curve, it is found that the generation loss tends to stabilize convergence, which is in line with the expectation of accurate classification. The discrimination loss gradually rises from the low point of the curve to about 0.07. This means that the modality classifier is difficult to distinguish the common patterns, and the specific patterns are linearly independent of the common patterns. As shown in Fig. 5, the total loss tends to convergence with the adversarial max–min game.

In addition, Fig. 6 illustrates the intermediate feature visualization of different modules on the Augsburg dataset; thus, the effectiveness is exhibited intuitively. Specifically, we randomly select proportional features for visualization, including original

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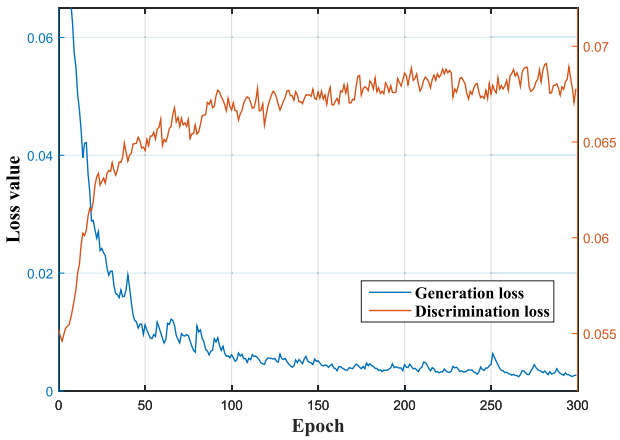

Fig. 4. Curve of generation loss $\mathcal{L}_{\text{gen}}$ and discrimination loss $\mathcal{L}_{\text{dis}}$.

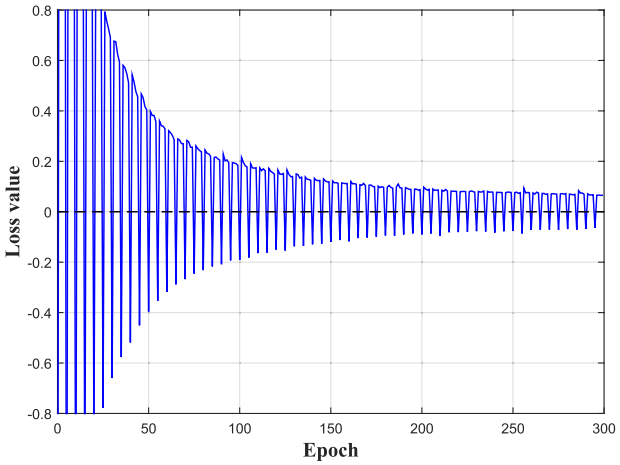

Fig. 5. Curve of total loss $\mathcal{L}$.

features, common patterns, common and specific patterns, as well as output features of PSM. From Fig. 6(a) and (b), it is observed that the original features of different classes are poorly distinguishable, and the common patterns as basic representation still limited the feature discrimination. In this article, the common patterns and specific patterns are integrated to generate the complementary feature representation, which significantly improved feature discrimination compared with the merely common pattern, as shown in Fig. 6(c). Fig. 6(d) indicates that the PSM further eliminates feature redundancy; thus, the features of the same classes are closer to each other, while a large margin separates features in different classes, which indicates the effectiveness of the proposed framework.

III. Experimental Results and Discussion

In this section, the description of experimental data and configuration were clearly stated, and then, the comprehensive experiments are conducted to validate the effectiveness of the proposed ACL-CNN as well as each of its components.

A. Experimental Data

The Berlin dataset was taken by the HyMap hyperspectral sensor in the Berlin area with a spatial resolution of 30 m,

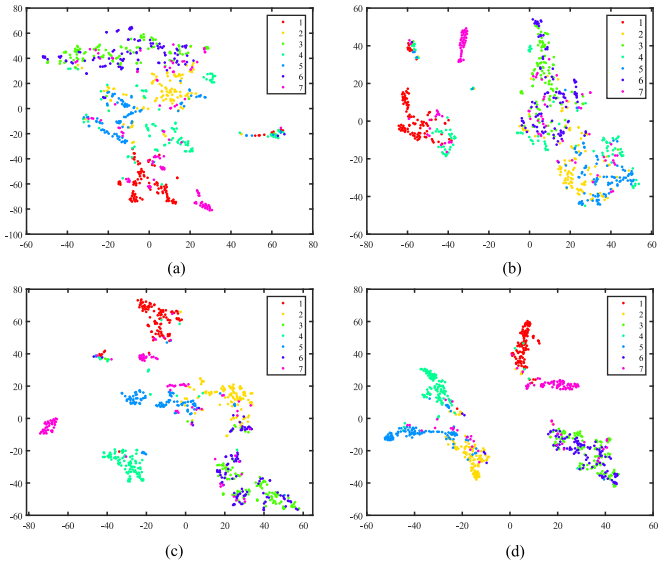

Fig. 6. Intermediate feature visualization of different modules in the proposed ACL-CNN. (a) Original features. (b) Common patterns. (c) Common and specific patterns. (d) Outputs of PSM.

which can be freely downloaded on website.[1] The HyMap hyperspectral sensor records the spectral information of land covers from visible light to infrared in the wavelength range of 400–2500 nm. The SAR image corresponding to the geographical location is the dual polarization (VV–VH) single look complex (SLC) of the Sentinel-1 satellite downloaded from European Space Agency (ESA). The SAR data are input into the preprocessing workflow based on SNAP toolbox,[2] including orbit correction, radiometric correction, penetration, and so on. The Berlin scene consists of 1723×476 pixels.

The Augsburg dataset contains HS, SAR, and digital surface model (DSM) images, which are captured by the HySpex sensor, C-band SAR sensor of Sentinel-1 satellite, and DLR-3K system, respectively. The HSI has 180 spectral bands ranging from 400 to 2500 nm. The DSM image and SAR image provide elevation and polarization information of land covers, respectively. There are 332×485 pixels in the Augsburg scene.

The Yellow River Estuary dataset is composed of HS, SAR, and MS images, which provide abundant spectral, polarization, and texture information, respectively. The HSI and MSI were taken by ZiYuan1-02D satellite in the Yellow River estuary in Shandong Province. The spatial resolution of HSI and MSI is 30 and 10 m, respectively. The number of bands of HSI is 166, including 76 bands of near infrared and 90 bands of short wave infrared. The SAR image corresponding to geographical locations was captured by the Sentinel-1 satellite. The Yellow River Estuary scene comprises 960×1170 pixels.

The pseudo-color images of the three datasets are shown in Fig. 7. The ground-truth (GT) maps of the Berlin and Augsburg datasets are generated by using the OpenStreetMap data [51]. The Yellow River Estuary dataset is annotated during the field investigation. The number of training and testing samples is listed in Table I. In addition, data augmentation is employed in the training stage. Specifically, a group of random seeds is generated for rotation, flipping, and noise. Therefore,

[1] http://doi.org/10.5880/enmap.2016.002
[2] https://step.esa.int/main/toolboxes/snap/

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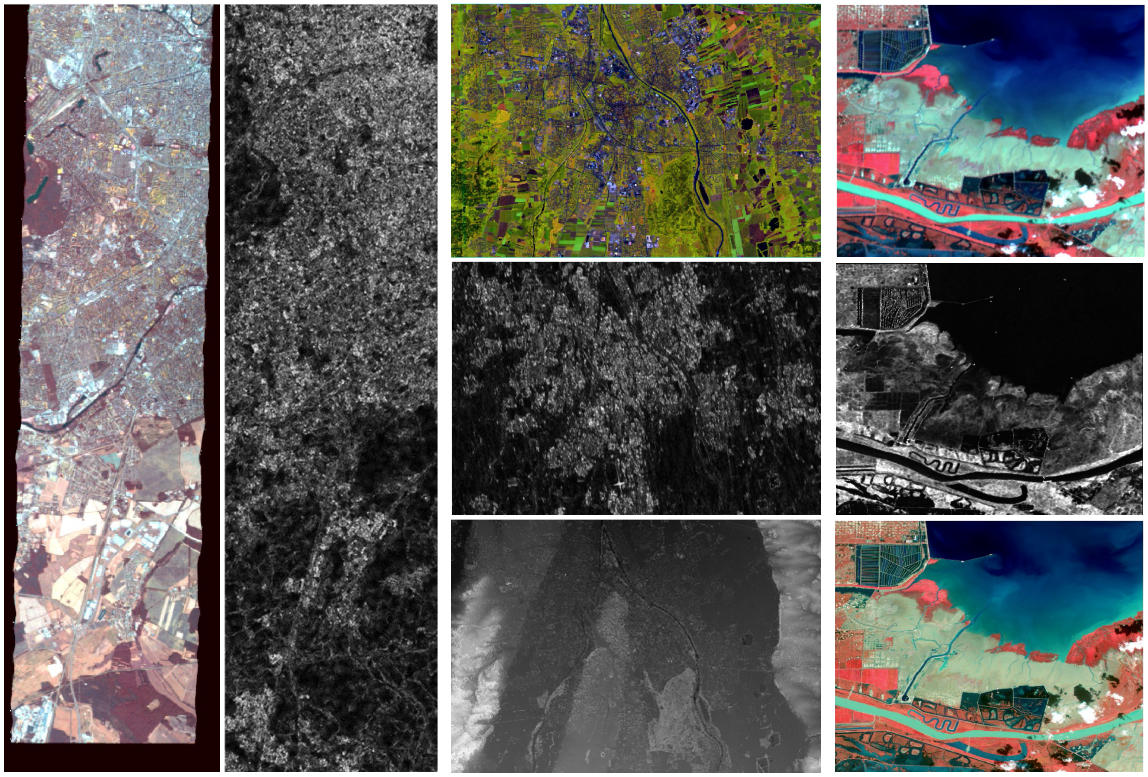

(a) (b) (c)

Fig. 7. Multisource remote sensing datasets. (a) Berlin dataset (HSI + SAR). (b) Augsburg dataset (HSI + SAR + DSM). (c) Yellow River Estuary dataset (HSI + SAR + MSI).

TABLE I

NUMBER OF TRAINING AND TESTING SAMPLES OF THREE MULTISOURCE DATASETS

No.	Berlin dataset			Augsburg dataset			Yellow River Estuary dataset		
	Class	Training	Testing	Class	Training	Testing	Class	Training	Testing
1	Forest	443	54511	Forest	146	13361	Spartina Alterniflora	735	39784
2	Residential Area	423	268219	Residential Area	264	30065	Suaeda Salsa	2519	118213
3	Industrial Area	499	19067	Industrial Area	21	3830	Tamarix Forest	1297	35216
4	Low Plants	376	58906	Low Plants	248	26609	Tidal Creek	529	15673
5	Soil	331	17095	Allotment	52	523	Mudflat	702	24592
6	Allotment	280	13025	Commercial Area	7	1638			
7	Commercial Area	298	24526	Water	23	1507			
8	Water	170	6502						
	Total	2820	461851	Total	761	77533	Total	5782	233478

the number of training samples is expanded several times. Here, 80% of the training samples are utilized for training, and the remaining 20% are used for validation.

B. Experimental Settings

Three groups of experiments are designed to evaluate the performance of the proposed method in the task of multisource remote sensing image classification.

1) The classification results are measured for three datasets, considering different neighborhood sizes r, the hyper-parameter of discrimination loss λ, and the depth of encoder networks n. Besides, the effectiveness of components in the proposed ACL-CNN modules is verified through ablation experiments. The candidate sets are defined as $[r = 7, 9, 11, 13, 15, 17]$, $[\lambda = 0.001, 0.005, 0.01, 0.1]$, and $[n = 2, 3, 4, 5]$.
2) For comparison, several related technologies are utilized to classify the multisource data, including SVM [52], local binary patterns and ELM (LBPs-ELM) [53], shared and specific feature learning (S2FL) [25], two-branch CNN (TBCNN) [36], coupled CNN (CPCNN) [37], and depthwise feature interaction network (DFINet) [40]. Especially, the SVM and LBP-ELM are conducted on the stacked the multisource images. Other approaches achieve the information integration through feature fusion.
3) The last experiment is implemented to further analyze the robustness of the proposed method. Concretely, the number of training samples based on a candidate set $[20\%, 40\%, 60\%, 80\%]$ is utilized to verify the classification performance in few-shot scene. In addition, the proposed method is expansively deployed in the collaborative classification scene of three data sources.

The overall accuracy (OA) is mainly utilized to evaluate the performance. Besides, average accuracy (AA), class accuracy (CA), and kappa coefficient (Kappa) are considered for

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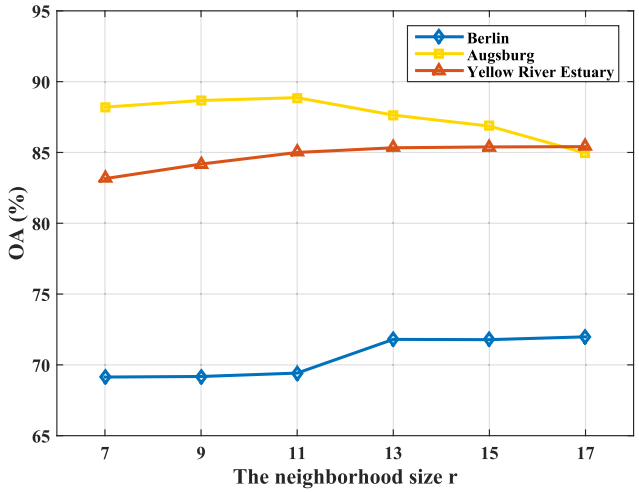

Fig. 8. Relationship between the OA and the neighborhood size.

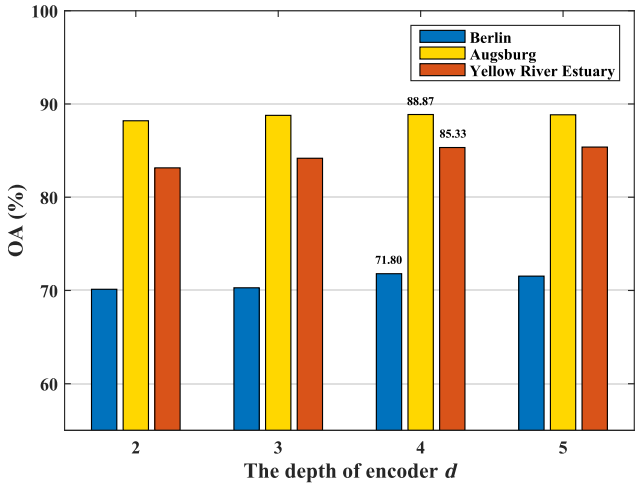

Fig. 9. Relationship between the OA and the depth of encoder.

TABLE II
RELATIONSHIP BETWEEN THE OA AND THE WEIGHT HYPERPARAMETER

Method	Metric	Dataset		
		Berlin	Augsburg	Yellow River Estuary
λ=0.001	OA (%)	**71.80**	**88.87**	**85.33**
λ=0.002		71.25	88.59	85.32
λ=0.005		70.00	88.42	84.93
λ=0.01		—	87.89	83.42

TABLE III
ABLATION EXPERIMENT OF THE PROPOSED ACL-CNN

Index	Method	Metric	Dataset		
			Berlin	Augsburg	Yellow River Estuary
1	Concat	OA (%)	67.45	87.34	82.12
2	SEM		68.11	87.75	83.24
3	CAM		69.85	88.50	84.26
4	PSM		**71.80**	**88.87**	**85.33**
5	PSM*	OA (%)	64.15	85.84	81.30

quantitative analysis. For experiment 2), the classification maps are generated to evaluate the performance of some typical areas.

C. Experiments 1: Analysis on Parameters

1) Analysis of the Neighborhood Size: For pixelwise classification, the neighborhood information is obtained through the image patch of the central pixel to enhance the feature representation. The relationship between OA and different neighborhood sizes r is shown in Fig. 8. Here, the neighborhood sizes are selected from the candidate set $[r = 7, 9, 11, 13, 15, 17]$. For the Berlin and Yellow River Estuary datasets, when r increases from 7 to 13, the OA values constantly increase. When $r = 13$, the OA value is close to the maximum value, increasing by more than 1%. Because the larger neighborhood size dilutes the central pixel, the OA value increases slightly or even decreases, as r increases to greater than 13. Thus, $r = 13$ is a reasonable choice. For the Augsburg dataset, the OA value reaches the maximum at $r = 11$, so 11 is selected as the optimal neighborhood size.

2) Analysis of the Weight Hyperparameter: For the ACL, the discrimination loss is equipped to distinguish modality labels and feature patterns. A hyperparameter λ is defined to balance the contribution of the generation loss and discrimination loss. λ is empirically set to $[0.001, 0.002, 0.005, 0.01]$. The relationship between the OA and the weight hyperparameter is listed in Table II. As the weight of discrimination loss increases to 0.005, the classification performance deteriorated gradually. When $\lambda = 0.001$, the proposed ACL-CNN achieves promising performance on three datasets.

3) Analysis of the Depth of Encoder: The common/specific encoder is composed of several convolutional layers in series. However, adding the number of convolutional layers introduces a large number of parameters that need to be optimized. This puts forward higher requirements for the amount of training set. Next, we discuss the impact of the number of convolutional layers on OA. As shown in Fig. 9, the encoder with a depth of 2 cannot meet the needs of accurate classification. That is to say, the high-level semantic features rely on deeper encoders. It is found that the encoder with a depth of 4 considers model optimization and feature embedding.

4) Ablation Experiment: In this section, the components of the proposed ACL-CNN are discussed by the ablation experiment, including the discriminator and PSM.

a) Pattern sampling module: Indexes 1–4 in Table III describe the effectiveness of different fusion methods. Specifically, several attention modules are chosen for performance verification including the concatenate (Concat) operator, squeeze-and-excitation module (SEM) [54], CAM [55], and PSM. When Concat is employed for integrating specific patterns, the precision improvement is obviously limited. This is because the noise elements in the specific patterns reduce the gain. The SEM denotes that the channel attention is implemented on the features of Concat. Therefore, the classification precision is further improved. The CAM emphasizes meaningful information through attention weight to obtain competitive performance. In contrast, the PSM achieves superior performance, and the PSM is easily extended to other scenarios with more modalities.

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TABLE IV
CLASS-SPECIFIC CLASSIFICATION ACCURACY (%) USING DIFFERENT METHODS ON THE BERLIN DATASET

No.	Class	The Classification Performance						
		SVM	**LBP-ELM**	**S2FL**	**Two-branch CNN**	**CPCNN**	**DFINet**	**ACL-CNN**
1	Forest	68.11	**85.80**	79.52	81.14	78.35	80.29	78.29
2	Residential Area	62.22	36.31	49.41	61.15	60.29	61.93	**69.71**
3	Industrial Area	29.01	46.08	45.18	57.68	61.21	47.44	**65.26**
4	Low Plants	84.93	83.67	70.50	84.03	75.67	80.01	**86.74**
5	Soil	85.99	**90.65**	81.47	79.80	85.69	77.54	82.37
6	Allotment	44.12	0.00	61.31	70.81	**75.62**	73.42	75.01
7	Commercial Area	31.75	0.27	29.63	34.17	37.16	**49.11**	37.74
8	Water	66.06	49.63	57.24	78.53	71.52	77.59	**81.71**
OA (%)		63.25	47.86	56.06	66.06	64.72	66.24	**71.80**
AA (%)		59.02	49.05	59.28	68.41	68.19	68.42	**72.10**
Kappa		0.4862	0.3424	0.4246	0.5393	0.5242	0.5398	**0.6034**

b) Discriminator: Index 5 in Table III reports the relationship between the OA and the discriminator. PSM* means that the common and specific patterns are extracted without discriminator. It is observed that the discriminator effectively filters out the common and specific patterns from multisource data, which promote the improvement of classification accuracy by at least 7.65%, 3.03%, and 4.03% on three datasets, respectively. It confirms that the common patterns and specific patterns extracted by ACL are more discriminative.

D. Experiments 2: Classification Performance and Analysis

This section focuses on validating the performance of the proposed ACL-CNN with respect to some of the most well-known classification techniques. The classification indicators involved in Section III-B are reported through tables. For qualitative analysis, the classification maps generated by methods with optimal and suboptimal performance are exhibited.

1) Results on the Berlin Dataset: Focusing on the Berlin scene, it is possible to observe from Table IV that the proposed method achieves a consistent improvement in OA, AA, and Kappa, which is mainly due to the strong recognition ability of building areas (classes 2, 3, and 7). Among all the competitors, the CNN-based methods outperform the SVM and LBP-ELM. This is expected, as these methods automatically joint multisource features, which provide the more discriminative information than the concatenation in SVM and LBP-ELM. The DFINet achieves the second highest OA value among several CNN-based methods. However, the OA value is still 5.56% lower than that of the proposed method. In addition, the classification map in Fig. 10 of DFINet tends to be rather noisy, mainly because some industrial area and residential area are misclassified into commercial area. The proposed method is able to generate a clear classification map, which is closer to the real distribution of land covers.

2) Results on the Augsburg Dataset: Focusing on the Augsburg scene, the classification maps are illustrated in Fig. 11, while the CA, OA, AA, and Kappa values are listed in Table V. As a general comment, the proposed ACL-CNN achieves the remarkable improvements in the global metrics. For instance, the proposed method consistently outperforms the other competitors and improves by at least 1.3%, 0.3%, and 0.0206% in OA, AA, and Kappa. As listed in Table V, the spectral classifiers (SVM and LBP-ELM) fail to obtain satisfactory results, which can be overcome by the learnable feature fusion. The feature fusion methods achieve relatively close performance, while the proposed method is still excellent

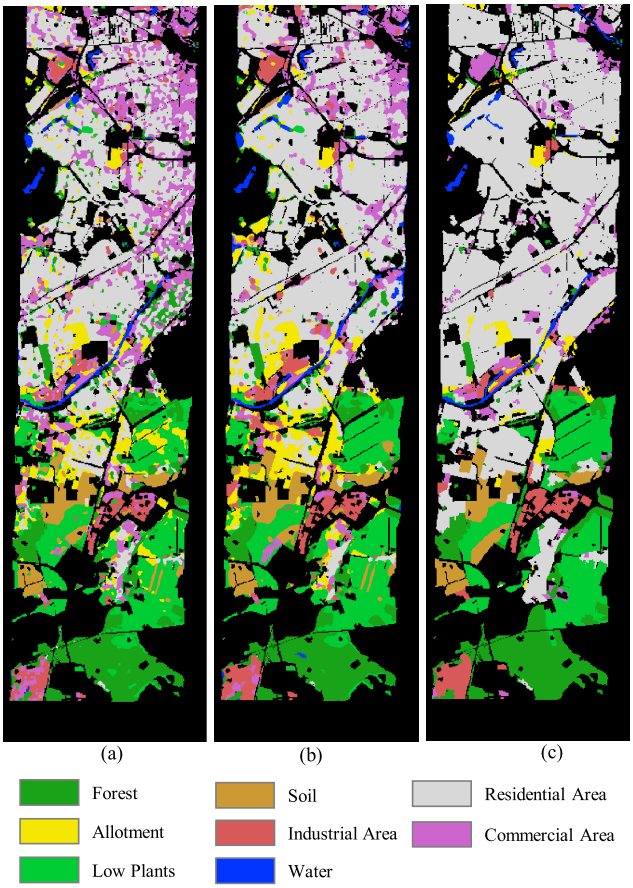

Fig. 10. Classification maps for the Berlin dataset using two methods with optimal and suboptimal performance. (a) Classification map of DFINet (OA = 66.24%). (b) Classification map of ACL-CNN (OA = 71.80%). (c) GT map.

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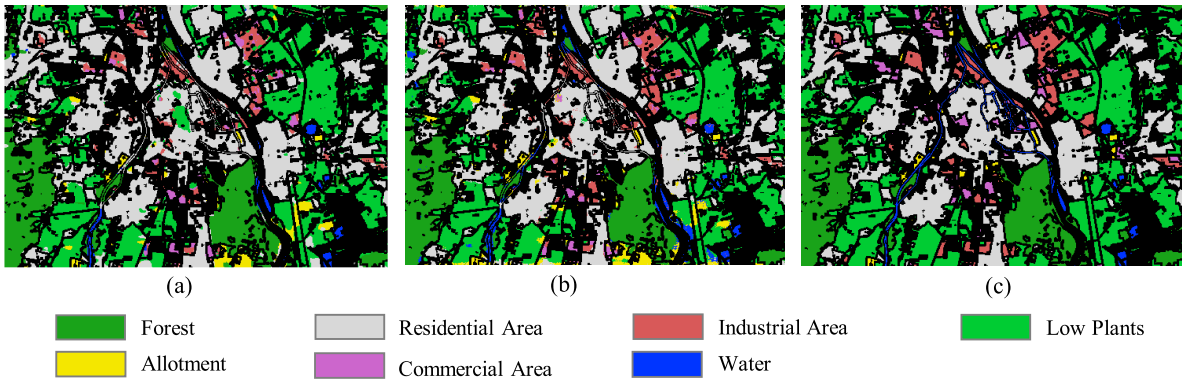

Fig. 11. Classification maps for the Augsburg dataset using two methods with optimal and suboptimal performance. (a) Classification map of DFINet (OA = 87.57%). (b) Classification map of ACL-CNN (OA = 88.87%). (c) GT map.

TABLE V
CLASS-SPECIFIC CLASSIFICATION ACCURACY (%) USING DIFFERENT METHODS ON THE AUGSBURG DATASET

No.	Class	The Classification Performance						
		SVM	**LBP-ELM**	**S2FL**	**Two-branch CNN**	**CPCNN**	**DFINet**	**CL-CNN**
1	Forest	90.33	87.12	88.80	93.96	93.14	**96.28**	94.75
2	Residential Area	90.47	83.79	86.36	95.19	96.46	96.50	**97.05**
3	Industrial Area	20.37	32.61	38.90	58.93	52.30	53.42	**66.55**
4	Low Plants	84.57	70.50	**90.53**	85.01	88.61	85.05	87.04
5	Allotment	36.71	57.36	**68.64**	50.86	55.64	54.30	63.86
6	Commercial Area	9.58	10.74	8.97	**27.35**	3.24	21.12	17.09
7	Water	45.65	44.99	47.45	48.37	22.30	46.91	**49.30**
OA (%)		82.01	74.80	86.36	87.05	87.33	87.57	**88.87**
AA (%)		53.95	55.30	61.38	67.65	58.81	64.80	**67.95**
Kappa		0.7374	0.6421	0.7626	0.8160	0.8163	0.8214	**0.8420**

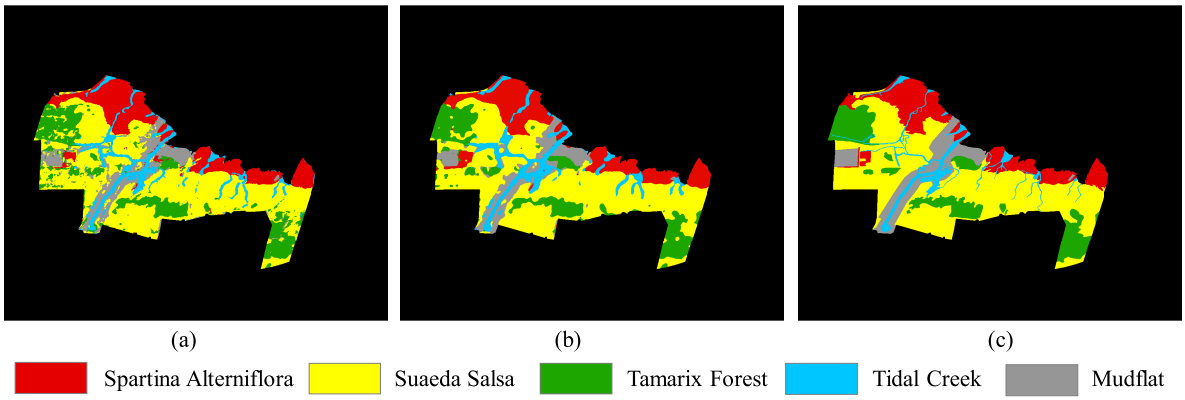

Fig. 12. Classification maps for the Yellow River Estuary dataset using two methods with optimal and suboptimal performance. (a) Classification map of CPCNN (OA = 82.95%). (b) Classification map of ACL-CNN (OA = 85.33%). (c) GT map.

in distinguishing building areas. As shown in Fig. 11, the residential area and industrial area can be identified more accurately, thus resulting in a better border delineation.

3) Results on the Yellow River Estuary Dataset: Focusing on the Yellow River Estuary scene, the proposed method introduces around 2.5% points of improvement in OA when compared to method with suboptimal performance. It is observed from Table VI that the proposed ACL-CNN obtains higher accuracy in the complex areas with Suaeda salsa and Tamarix forest (classes 2 and 3). For the SVM and LBP-ELM, the OA values have little difference with that of ACL-CNN. This is because the accuracy of Tamarix forest (class 3) is relative low, resulting in low AA values. As shown in Fig. 12(a), the CPCNN with the second highest OA values suffers from the artifacts and fragmentation, so that generating a visually noisy classification map. In contrast, the classification map generated by the proposed ACL-CNN has fewer wrongly labeled pixels. Besides, as shown in Fig. 7(c), some areas in the HSI are covered by clouds. In contrast, the corresponding SAR image has the ability to generate observations regardless of the weather conditions. Therefore, the land-covers classification of cloud-obscured areas is achieved by adaptively strengthening the complementary representation from SAR images.

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TABLE VI
CLASS-SPECIFIC CLASSIFICATION ACCURACY (%) USING DIFFERENT METHODS ON THE YELLOW RIVER ESTUARY DATASET

No.	Class	The Classification Performance						
		SVM	**LBP-ELM**	**S2FL**	**Two-branch CNN**	**CPCNN**	**DFINet**	**ACL-CNN**
1	Spartina Alterniflora	91.18	91.16	83.48	**92.74**	92.21	92.15	92.67
2	Suaeda Salsa	81.62	85.94	77.03	83.07	83.49	84.82	**86.90**
3	Tamarix Forest	74.26	73.64	68.04	74.98	81.60	81.18	**83.71**
4	Tidal Creek	79.37	69.60	67.47	**80.22**	78.52	76.67	79.04
5	Mudflat	70.57	68.30	45.56	**72.47**	70.16	65.21	72.21
OA (%)		80.82	82.02	72.82	82.19	82.95	82.91	**85.33**
AA (%)		79.40	77.73	68.31	80.70	81.19	80.01	**82.91**
Kappa		0.7233	0.7346	0.6028	0.7429	0.7534	0.7517	**0.7865**

Generally speaking, the proposed method achieves the best performance in both quantitative and qualitative analyses compared with other competitors. The comparisons also demonstrate that the ACL strategy has the ability to filter out different patterns to encourage the feature discrimination.

E. Experiments 3: Analysis on Performance Expansion

In practical application, most land-covers classification technologies have exposed its limitation in the scenario of diverse data sources or insufficient labels. Therefore, the performance expansion of the proposed method is further analyzed in this section.

1) Analysis of the Performance Using Three Modalities: The Augsburg dataset has an additional DSM image with abundant elevation information. The MSI of the Yellow River Estuary dataset provides the complementary texture information. Therefore, the proposed ACL-CNN is further implemented on three modalities of remote sensing data. The modality labels are redefined as [1, 0, 0], [0, 1, 0], and [0, 0, 1], respectively. The discrimination loss and generation loss are redefined as follows:

$$\begin{cases} \mathcal{L}_{cf} = -\dfrac{1}{N}\displaystyle\sum_{j=1}^{N}\big(y_m \cdot \big(\log p\big(\mathbf{F}_c^1, \Theta_D\big) + \log p\big(\mathbf{F}_c^2, \Theta_D\big) \\ \qquad + \log p\big(\mathbf{F}_c^3, \Theta_D\big)\big)\big) \\ \mathcal{L}_{sf} = \dfrac{1}{N}\displaystyle\sum_{j=1}^{N}\big(\|\mathbf{F}_c^1 - \mathbf{F}_s^1\|_2 + \|\mathbf{F}_c^2 - \mathbf{F}_s^2\|_2 + \|\mathbf{F}_c^3 - \mathbf{F}_s^3\|_2\big) \end{cases} \tag{17}$$

$$\begin{cases} \mathcal{L}_{re} = \dfrac{1}{N}\displaystyle\sum_{j=1}^{N}(\|\hat{\mathbf{F}}^1 - \mathbf{F}^1\|_2 + \|\hat{\mathbf{F}}^2 - \mathbf{F}^2\|_2 + \|\hat{\mathbf{F}}^3 - \mathbf{F}^3\|_2 \\ \mathcal{L}_{ce} = -\dfrac{1}{N}\displaystyle\sum_{j=1}^{N} y \cdot \log p(\mathbf{F}, \Theta_C). \end{cases} \tag{18}$$

The results on three modalities are reported in Table VII. The comparison methods (SVM and LBP-ELM) receive the three modalities by concatenating the HSI, SAR image, and DSM/MS images. It is observed that the SVM and LBP-ELM slightly improve the performance, while the proposed ACL-CNN reaches a consistent OA improvement of 2.13% and 1.32% using three modalities. Specifically, for the Augsburg scene, the DSM image with elevation information significantly improves the classification accuracy of class 4 (low plants) and class 6 (commercial area) by 6.06% and 6.90%, respectively. For the Yellow River Estuary scene, the accuracies of class 4 (tidal creek) and class 5 (mudflat) are greatly improved. This shows that texture information is beneficial to the recognition of nonvegetation types. In general, the proposed ACL-CNN is easily extended to the classification tasks with more modalities and, thus, further enhances the extraction of complementary information.

TABLE VII
CLASS-SPECIFIC CLASSIFICATION ACCURACY (%) USING DIFFERENT METHODS ON THE AUGSBURG AND YELLOW RIVER ESTUARY DATASETS WITH THREE MODALITIES

Augsburg dataset						
No.	**SVM**		**LBP-ELM**		**ACL-CNN**	
	m=2	m=3	m=2	m=3	m=2	m=3
1	90.33	90.67	87.12	87.64	**94.75**	93.72
2	90.47	90.60	83.79	83.88	97.05	**98.40**
3	20.37	20.44	32.61	32.58	**66.55**	58.72
4	84.57	84.50	70.50	70.57	87.04	**93.10**
5	36.71	36.14	57.36	57.36	**63.86**	**63.86**
6	9.58	9.58	10.74	10.74	17.09	**23.99**
7	45.65	45.65	44.99	45.19	49.30	46.72
OA (%)	82.01	82.10	74.80	74.95	88.87	**91.00**
AA (%)	53.95	53.94	55.30	55.42	67.95	**68.36**
Kappa	0.7374	0.7386	0.6421	0.6442	0.8420	**0.8709**

Yellow River Estuary dataset						
No.	**SVM**		**LBP-ELM**		**ACL-CNN**	
	m=2	m=3	m=2	m=3	m=2	m=3
1	91.18	94.62	91.16	91.20	92.67	92.21
2	81.62	82.62	85.94	89.73	86.90	87.84
3	74.26	73.39	73.64	66.13	83.71	83.35
4	79.37	78.13	69.60	66.36	79.04	80.88
5	70.57	72.48	68.30	60.47	72.21	77.46
OA (%)	80.82	81.90	82.02	81.77	85.33	**86.65**
AA (%)	79.40	80.25	77.73	74.78	82.91	**84.75**
Kappa	0.7233	0.7375	0.7346	0.7238	0.7865	**0.8062**

2) Analysis of the Relationship Between the OA and the Number of Training Samples: The network model based on CNN needs a large number of labeled samples for optimization. However, field investigation is indispensable in the task of remote sensing image annotation, which is time-consuming and laborious. As shown in Fig. 13, training samples with

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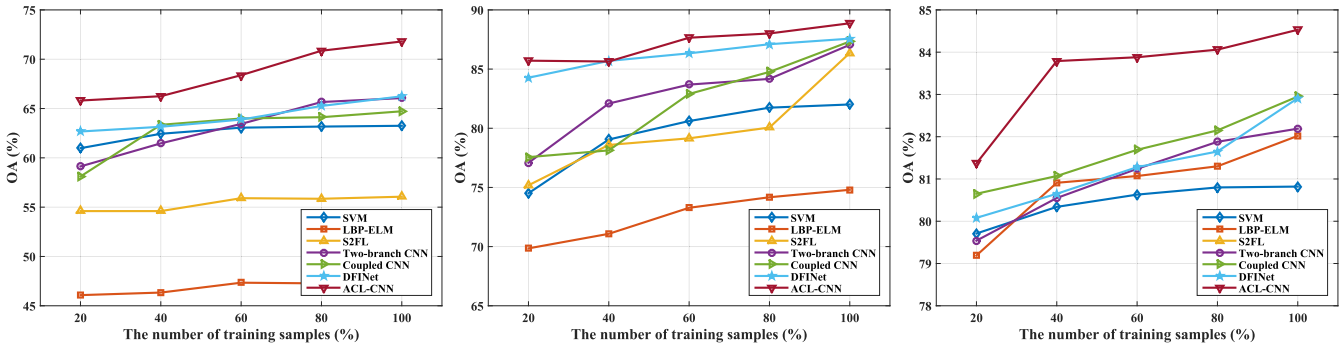

Fig. 13. Relationship between the OA and the number of training samples on three datasets. (a) Result on the Berlin dataset. (b) Result on the Augsburg dataset. (c) Result on the Yellow River Estuary dataset.

different proportions [20%, 40%, 60%, 80%] are randomly selected for model validation. For better analysis, the curves of some comparison methods are not exhibited. In general, the proposed ACL-CNN is able to maintain the relatively superior OA values compared with other competitors. For the CNN-based method, when the proportion of training samples is less than 60%, the classification performance is significantly reduced, especially for TBCNN. This is because compared with TBCNN, the CPCNN reduces model parameters through model coupling, and the DFINet guides biased estimation through multiple loss functions, which reduces the dependence on training samples. Similar to DFINet, the proposed ACL-CNN is jointly optimized by multiple loss functions. Therefore, the solution is regularized into a smaller range, which promotes the performance in few-shot scene.

IV. Conclusion

This article proposes an ACL-CNN with the aim of improving classification performance. In general, the proposed ACL-CNN benefits from two aspects. One is the adversarial max–min game to filter out the common patterns and specific patterns of multisource data. The common patterns have the ability to basically describe the characteristics of land covers. The specific patterns that are linearly independent of common patterns provide supplementary information. Therefore, the compact and discriminative features are generated. The other one is the PSM to eliminate the singular noise. The mutual-exclusion relationship between specific patterns is extracted by the PSM. The obtained experimental results, conducted using three multisource datasets, demonstrate the significant improvements introduced by the proposed method. Thus, the meaningful information is retained, while the unnecessary one is suppressed. Besides, the proposed ACL-CNN provides a highly efficient mechanism to extend itself to the scene with more modalities.

Despite the high accuracy and flexibility provided by the proposed ACL-CNN, there are some unresolved issues that may present challenges over time. In particular, our future work will be aimed at the following directions: 1) adaptive allocation of weight parameters of loss functions and 2) introducing advanced data augmentation or migration schemes to improve model robustness of few-shot scenarios.

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